INCH-POUND
MIL-DTL-24643/17G
27 November 2023
SUPERSEDING
MIL-DTL-24643/17F
1 October 2009

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +221 DEGREES FAHRENHEIT (+105 DEGREES CELSIUS), 1,000 VOLTS, TYPE LSFSGU

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction (Watertight with Circuit Integrity)

First	-	Copper conductor, coated or uncoated (see table I for size).
Second	-	Thermoset insulation, extruded or taped (see table I for thickness).
Third	-	An optional glass braid.
Fourth	-	An optional covering (unless required by 3.3.6 of MIL-DTL-24643).
Fifth	-	The four conductors shall be cabled together in accordance with 3.4.4 of MIL-DTL-24643. Fillers may be used as necessary to form a firm, well-rounded assembly. Standard identification code applied by method 1 or 3, or letter identification code applied by method 5. If identification method 1 is used, outer surface to be printed shall appear black. Cabling sequence shall be consecutive.
Sixth	-	An optional binder.
Seventh	-	Cross-linked polyolefin jacket. See table I for thickness.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

Distribution Statement A. Approved for public release: distribution is unlimited.

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TABLE I. Details.

Military part no. M24643/17	Type and size	Conductor size (AWG)	Cable overall diameter max. (inches)	Insulation thickness min. avg. (inch)	Jacket thickness min. avg. (inch)	Conductor resistance (ohms)	Insulation resistance (megohms)	Accelerated service loading (amperes)
-01UN	LSFSGU-3	16 (class B)	0.447	0.018	0.030	4.3	500	N/A
-02UN	LSFSGU-4	14 (class B)	0.513	0.018	0.040	2.68	500	N/A
-03UN	LSFSGU-9	10 (class B)	0.630	0.018	0.040	1.08	500	N/A
-04UN	LSFSGU-23	7 (class B)	0.890	0.018	0.050	0.543	500	115
-05UN	LSFSGU-50	3 (class C)	1.050	0.035	0.050	0.210	200	185
-06UN	LSFSGU-75	1 (class C)	1.240	0.035	0.050	0.134	175	250
-07UN	LSFSGU-100	0 (class D)	1.358	0.035	0.060	0.106	160	295
-08UN	LSFSGU-150	000 (class D)	1.625	0.040	0.060	0.0674	135	400
-09UN	LSFSGU-200	0000 (class D)	1.820	0.050	0.060	0.0530	125	465

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REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.)	See table I
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	
Sizes 3 through 9	3,000
Sizes 23 and larger	5,000
Insulation resistance (megohms/1,000 feet, min.)	
Conductor to conductor	See table I
Conductor continuity	No failure

Group A:

Visual and dimensional	No failure
Watertightness (see MIL-DTL-24643 for limits of water leakage)	No failure
Abrasion resistance (scrapes, min.) (sizes 40 and smaller) (extruded insulation only)	250
Crack resistance (sizes 3 through 100) (extruded insulation only)	No damage

Group B:

Cold bending (cable)	No failure
Thermoset proof test (percent, max.)	
Insulation (extruded insulation only)	50
Jacket (when tested at 392 °F [200 °C])	50
Drip (203±1.8 °F [95±1 °C]) (see MIL-DTL-24643 for limits of drip allowance)	No failure
Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1,300
Elongation (percent, min.)	160

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Tear (lb/in thickness, min.)	35

Group C:

Physicals (aged) (air oven)	
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (insulation, method 1 only) (cycles, min.)	25
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Cable fill (sealant) removability	No failure
Shrinkage	No failure

Group D:

Gas flame (circuit integrity) (1 hour)	No failure
Flame propagation (cable)	No failure

Qualification inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Cold working (-4±3.6 °F [-20±2 °C])	No damage
Gas flame (circuit integrity) (3 hours)	No failure
Accelerated service (sizes 23 and larger) (see table I for load current)	No failure
Aging and compatibility (cable) (257±9 °F [125±5 °C])	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50

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Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) – type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption (extruded insulation only)	No failure

UNIT ORDERING LENGTHS:

Type	Size	Feet (nominal)
LSFSGU	3	2,500
LSFSGU	4 and 9	2,000
LSFSGU	23 and 50	1,500
LSFSGU	75 through 150	1,000
LSFSGU	200	500

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2013-033)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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